

1. Administrative & Study Identification

Full Study Title: An Open-label, Multi-center, Phase I/II Study of ECI830 as a Single Agent and in Combination With Ribociclib and Endocrine Therapy in Patients With Advanced Hormone Receptor Positive, HER2-negative Breast Cancer and Advanced Solid Tumors
Short Title/Acronym: Study of ECI830 Single Agent or in Combination in Patients With Advanced HR+/HER2- Breast Cancer and Other Advanced Solid Tumors
Protocol Number: CECI830A12101 **NCT Number:** NCT06726148 **Posting ID:** 112657688834418496 **Trial Status:** RECRUITING (Currently recruiting participants) **Sponsor/Company Name:** Novartis Pharmaceuticals / Novartis **Investigational Product:** ECI830 (Single agent or in combination with ribociclib and fulvestrant) **Phase of Development:** Phase I/II (PHASE1, PHASE2) **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: * **Phase I:** Characterize the safety and tolerability of ECI830 as a single agent and in combination with ribociclib and fulvestrant, and identify the dose range for optimization/recommended dose for future studies.
* **Phase II:** Assess the anti-tumor activity of ECI830 in combination with ribociclib and fulvestrant in patients with hormone receptor-positive/human epidermal growth factor receptor 2-negative (HR+/HER2-) advanced breast cancer.

Secondary Objectives: To evaluate Pharmacokinetics (PK) including Area under the plasma concentration-time curve (AUC) and Maximum observed plasma concentration (Cmax), Overall Response Rate (ORR), Best Overall Response (BOR), Duration of Response (DoR), and Progression-Free Survival (PFS).

2.2 Background & Rationale To address the medical need for novel targeted therapies in patients with advanced HR+/HER2- breast cancer that has progressed on or following prior lines of hormone-based therapy and CDK4/6 inhibitors. ECI830, an investigational CDK2 inhibitor, aims to block the CDK2 enzyme that helps cancer grow and overcome treatment resistance, particularly in patients harboring CCNE1 amplification.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm escalation followed by expansion / Dose optimization. Single agent escalation may be followed by an expansion part

stratified by disease indication. **Blinding:** Open-label
Randomization: Escalation of the combination arm may continue into a randomized, open-label, Phase II with optional dose optimization in advanced breast cancer patients.

3.2 Planned Enrollment & Duration Target **\$N\$:** 280 participants
Number of Sites: Multi-center (including locations in the United States and Australia) **Estimated Study Start:** 03-Apr-2025
Estimated Primary Completion: 25-Sep-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years old. 2. **Phase I:** HR+/HER2- advanced breast cancer (aBC) with disease progression on or following at least one line of hormone-based therapy in combination with a CDK4/6 inhibitor and at least one additional line of systemic therapy for metastatic disease; OR Histologically and/or cytologically confirmed diagnosis of locally advanced or metastatic cancer with a CCNE1 amplification. *For dose expansion only:* no more than 3 prior lines of therapy for advanced or metastatic disease. 3. **Phase II:** HR+/HER2- aBC with disease progression on an aromatase inhibitor or tamoxifen in combination with a CDK4/6 inhibitor for unresectable/metastatic disease with no more than 2 lines of endocrine therapy. 4. Measurable disease as determined by RECIST v1.1. BC only: If no measurable disease is present, then at least one predominantly lytic bone lesion must be present that can be accurately assessed at baseline and is suitable for repeated assessment.

4.2 Exclusion Criteria 1. Previous treatment with a CDK2 inhibitor at any time. 2. Patients with inadequate bone marrow and/or organ functions with out-of-range laboratory values. 3. Clinically significant, uncontrolled heart disease and/or cardiac repolarization abnormality including MI, CABG, long QT syndrome, or risk factors for TdP. 4. Presence of symptomatic CNS metastases or CNS metastases that require local therapy or increasing doses of corticosteroids within 2 weeks prior to study entry. 5. For the combination treatment: Patients with symptomatic visceral disease or any disease burden that makes the patient ineligible for endocrine-based therapy. 6. For the combination treatment: Patients who could not tolerate the prescribed dose of ribociclib during a previous course of treatment, requiring dose reduction or permanent discontinuation due to adverse events. 7. For patients with BC: Patient is concurrently using hormone replacement therapy. 8. Women of childbearing potential (WOCBP) who are unwilling to use highly effective contraception methods, pregnant or nursing women.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: ECI830 administered continuously in specified dose cohorts, given alone or alongside standard doses of ribociclib and fulvestrant (Trade name: Faslodex, ATC code: L02BA03). **Route of Administration:** Oral (ECI830, ribociclib) and Intramuscular (fulvestrant). **Duration of Treatment:** Continuous treatment cycles until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for advanced oncology patients. **Prohibited:** Concurrent hormone replacement therapy (for BC patients), prior CDK2 inhibitors, and any therapies that significantly prolong the QT interval.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Scans (RECIST v1.1), Safety Labs
Baseline	Day 0	First Dose (ECI830 $\pm$ ribociclib/fulvestrant), Baseline PK evaluations
Interim	Every Cycle	Safety Labs, PK/PD Sampling, Tumor Imaging (every 8-12 weeks), AE Assessment
End of Study	At Progression	Final Vital Signs, Exit Interview, IP Accountability, Follow-up for Survival

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: * **Phase I:** * Incidence of Dose-Limiting Toxicities (DLTs) to determine the recommended dose. A DLT is defined as an adverse event or abnormal laboratory value of CTCAE grade ≥ 3 assessed as unrelated to disease, disease progression, inter-current illness or concomitant medications that occurs within the first 28 days of treatment. * Incidence of adverse events (AEs) and serious adverse events (SAEs) by treatment group, including changes in vital signs, ECGs, and laboratory results qualifying and reported as AEs. * Number of participants with dose interruptions, reductions and discontinuations (Assessment of tolerability). * **Phase II:** Anti-tumor activity measured by Overall Response Rate (ORR).

Measurement Method: CT/MRI scans evaluated per RECIST v1.1 guidelines; clinical safety assessments graded by NCI CTCAE.

7.2 Secondary & Exploratory Endpoints * **Phase I and II:** Area under the plasma concentration-time curve (AUC) of ECI830 and ribociclib * **Phase I and II:** Maximum observed plasma concentration (C_{max}) of ECI830 and ribociclib * **Phase I and II:** Best overall response (BOR) per RECIST v1.1 * Progression-Free Survival (PFS) * Duration of Response (DoR)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at each study visit using NCI CTCAE grading criteria, specifically monitoring for cardiac repolarization (QT interval) and bone marrow functions. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor and applicable regulatory bodies. **Data Monitoring Committee (DMC):** Internal Safety Review Committee responsible for dose-escalation decisions.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy (ORR, BOR, PFS); Safety Set for all AE/SAE and DLT evaluations. **Sample Size Justification:** N=280, powered to detect safety profiles in Phase I dose escalation and preliminary anti-tumor activity in Phase II expansion cohorts. **Handling of Missing Data:** Standard right-censoring methods for time-to-event endpoints (PFS, DoR) at the last adequate disease assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring to ensure protocol compliance and data integrity. **Audit Trail:** eCRF locking, source data verification, and data clarification forms tracking.

11. Study Identifiers & Governance

Primary ID: CECI830A12101 **Registry Number:** NCT06726148 **Posting ID:** 112657688834418496 **Sponsor/Lead Organization:** Novartis Pharmaceuticals / Novartis **Study Title:** Study of ECI830 Single Agent or in Combination in Patients With Advanced HR+/HER2- Breast Cancer and Other Advanced Solid Tumors **Official Regulatory Title:** An Open-label, Multi-center, Phase I/II Study of ECI830 as a Single Agent and in Combination With Ribociclib and Endocrine Therapy in Patients With Advanced Hormone Receptor Positive, HER2-negative Breast Cancer and Advanced Solid Tumors

12. Clinical Framework (Oncology Specific)

Disease Areas: Advanced HR+/HER2- Breast Cancer, Advanced CCNE1-amplified Solid Tumors **Preferred Terminology (UMLS/SNOMED):** Malignant neoplasm of breast (Breast Cancer), Advanced solid tumor **Diagnosis Threshold:** Histologically/cytologically confirmed advanced/metastatic disease; prior CDK4/6i progression or CCNE1

amplification **Medical Methodology:** CDK2 Inhibitor Targeted Therapy (ECI830) combined with Endocrine Therapy (fulvestrant/ribociclib) **Optimization Target:** Identification of optimal safe dosing and objective tumor response

13. Study Procedures & Methodology

Management Pathway: Advanced Breast Cancer progressive disease clinical pathway **Education Protocol (HCP):** Investigator meetings and site initiation training on ECI830 DLT management and RECIST evaluation **Education Protocol (Patient):** Informed consent process, reporting of AEs, and oral drug dosing diaries **Quality Audit Mechanism:** Centralized imaging review and rigorous onsite source data verification

14. Observation & Follow-Up Schedule

Total Duration: Estimated 41 months (Start: Apr 2025 – Completion: Sep 2028) **Onsite Visit Cadence:** Cycle 1 Day 1, and subsequent Day 1 of each treatment cycle **Remote Monitoring Frequency:** Continuous sponsor-level data review **Checklist Review Interval:** Prior to every cycle dispensing and DLT evaluation window

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				